# Role Standards and Competency Framework

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## 1. Engineering levels

All engineering job titles use the levels below. The experience ranges are guidelines for JDs and CV screening.

| Level | Title format | Experience | Scope |
|---|---|---|---|
| L1 | Associate Software Engineer | 0-1 year | Works on well-defined tasks with close guidance. |
| L2 | Software Engineer | 1-3 years | Delivers features independently within a team. |
| L3 | Senior Software Engineer | 3-6 years | Owns modules, reviews code, mentors L1/L2 engineers. |
| L4 | Lead Engineer | 6+ years | Leads a team of 4-8, owns technical design and delivery. |

Title examples: "Software Engineer (Backend)", "Senior Software Engineer (AI/ML)", "Lead Engineer (DevOps)".

## 2. Education

A BSc in Computer Science and Engineering (CSE), Electrical and Electronic Engineering (EEE), or a closely related field is **mandatory for L1 and L2**. For L3 and L4, equivalent professional experience may replace the degree requirement.

## 3. Mandatory and preferred skills by role family

The skills below apply from L2 upward. L1 candidates need only the fundamentals of the mandatory skills. For L3 and above, the extra items in the last column also become mandatory.

| Role family | Mandatory skills | Preferred skills | Extra mandatory for L3+ |
|---|---|---|---|
| Backend | Python or Java; REST API design; SQL with PostgreSQL; Git | FastAPI or Spring Boot; Docker; Redis; message queues | System design; code review experience |
| Frontend | JavaScript; React; HTML/CSS; Git | TypeScript; Next.js; Tailwind CSS; accessibility (WCAG) | Frontend architecture; performance tuning |
| AI/ML | Python; machine learning fundamentals; PyTorch or TensorFlow; SQL | LLM application development; RAG; LangChain or LangGraph; vector databases | Production deployment of at least one ML or LLM system |
| QA | Manual test design; API testing (Postman); bug tracking (Jira or Redmine) | Test automation (Playwright or Selenium); performance testing | Test strategy ownership |
| DevOps | Linux; Docker; CI/CD pipelines; Git | Kubernetes; AWS; Terraform; monitoring (Prometheus/Grafana) | On-call incident handling |

## 4. Behavioural competencies (all levels)

- Clear written and spoken English communication.
- Collaboration in agile teams (Scrum, two-week sprints).
- Ownership of quality, including writing tests for own code.

- Willingness to learn and share knowledge.